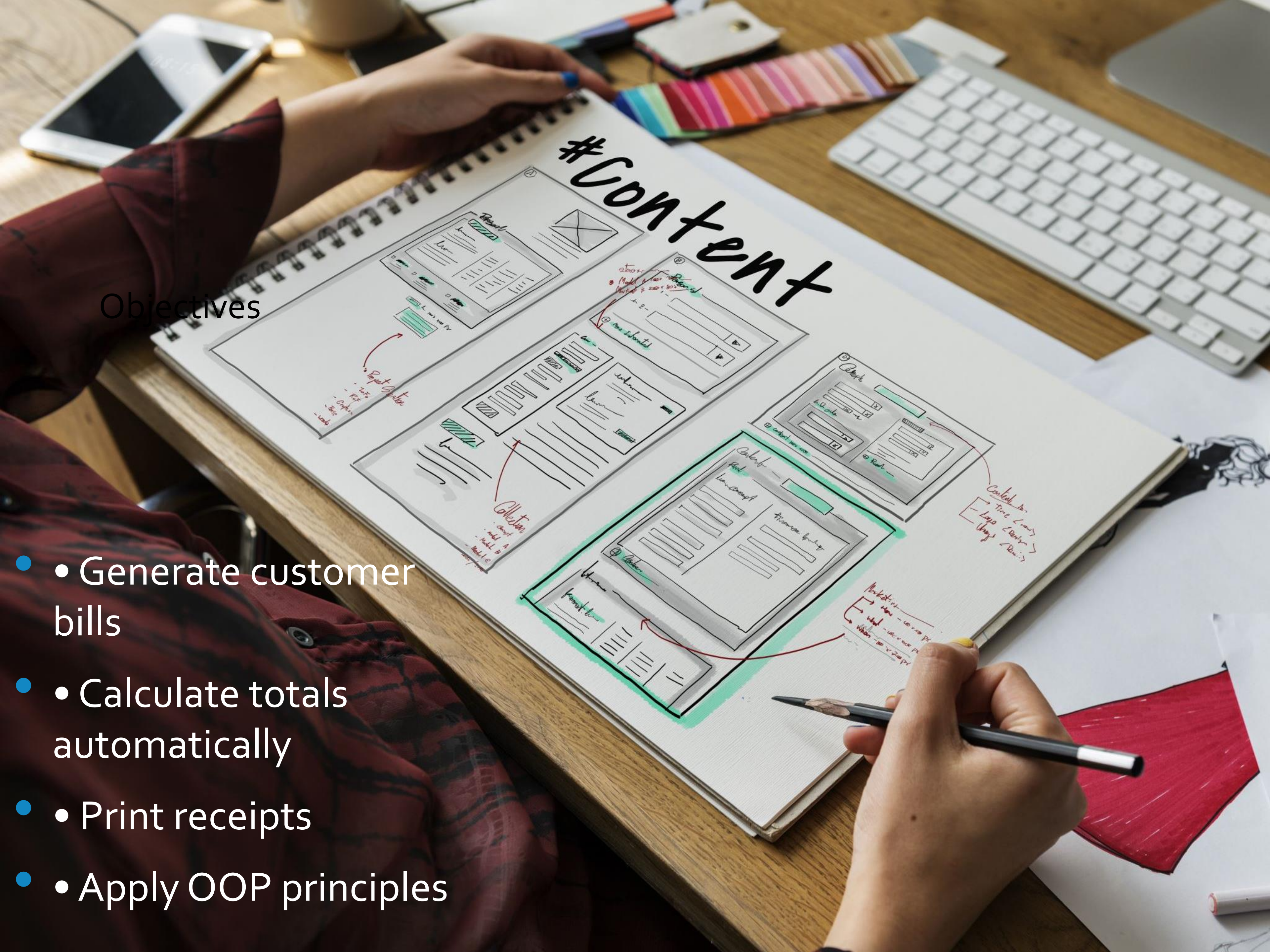
Design and Implementation of a Grocery Billing Receipt Generator using OOP

Project Presentation

Introduction

- A grocery billing system automates bill generation, item management, and receipt printing using Object-Oriented Programming concepts.



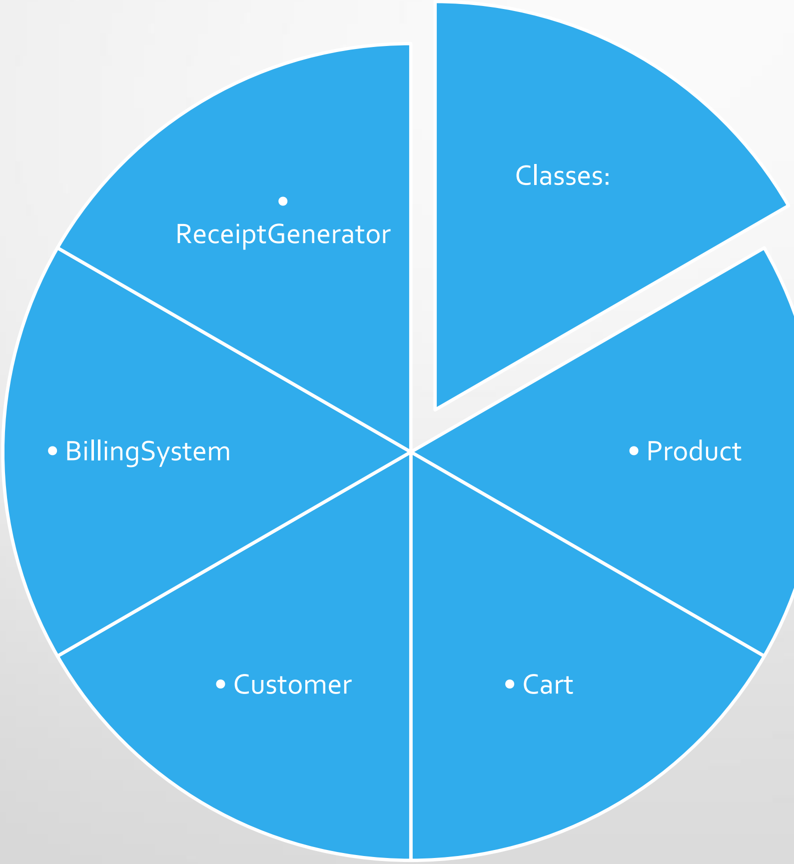
Objectives

- • Generate customer bills
- • Calculate totals automatically
- • Print receipts
- • Apply OOP principles

OOP Concepts Used

- • Class and Object
- • Encapsulation
- • Inheritance
- • Polymorphism
- • Abstraction

System Design



Implementation Flow

Add Products →
Add to Cart →
Calculate Total →
Generate Bill →
Print Receipt

Sample Features

- • Item name and price
- • Quantity handling
- • Tax calculation
- • Final receipt generation

JAVA CODE FOR PROGRAME

```
import java.util.Scanner;

class Product {
    private String name;
    private double price;
    private int quantity;

    public Product(String name, double price, int quantity) {
        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }

    public double getTotal() {
        return price * quantity;
    }

    public void display() {
        System.out.println(name + "\t" + price + "\t" + quantity + "\t" + getTotal());
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of items: ");
        int n = sc.nextInt();
        sc.nextLine();

        Product[] products = new Product[n];
        double grandTotal = 0;

        for (int i = 0; i < n; i++) {
            System.out.println("\nItem " + (i + 1));

            System.out.print("Enter Product Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Price: ");
            double price = sc.nextDouble();

            System.out.print("Enter Quantity: ");
            int qty = sc.nextInt();
            sc.nextLine();

            products[i] = new Product(name, price, qty);
            grandTotal += products[i].getTotal();
        }

        System.out.println("\n===== GROCERY BILL RECEIPT =====");
        System.out.println("Item\tPrice\tQty\tTotal");

        for (Product p : products) {
            p.display();
        }

        System.out.println("--------------------------------");
        System.out.println("Grand Total = Rs. " + grandTotal);
        System.out.println("Thank You for Shopping!");

        sc.close();
    }
}
```

CODE OUTPUT

Output

```
Enter number of items: 2

Item 1
Enter Product Name: RICE
Enter Price: 50
Enter Quantity: 2

Item 2
Enter Product Name: SUGAR
Enter Price: 80
Enter Quantity: 3

===== GROCERY BILL RECEIPT =====
Item    Price   Qty Total
RICE    50.0    2   100.0
SUGAR   80.0    3   240.0
--------------------------------
Grand Total = Rs. 340.0
Thank You for Shopping!

=== Code Execution Successful ===
```

Advantages

- Fast billing
- Reduced errors
- Easy maintenance
- Reusable code

Conclusion

- The Grocery Billing Receipt Generator demonstrates efficient use of OOP for developing a simple and scalable billing application.

Thank You